Brian Connor Liau

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1. Axis Guitar Project (IDEA Lab)

Embedded Hardware Engineer **Tools:** Altium Designer, KiCad, LTspice, ESP32, I2C, Lab Equipment

- Executed schematic capture and 2-layer PCB layout using KiCad, integrating an ESP32 microcontroller, IMU sensor, and power management circuits.
- Modeled power supply stability and analog front-end signals in LTspice prior to physical fabrication.
- Led hands-on board bring-up, debugging, and fine-pitch SMT component rework; used oscilloscopes and multimeters to analyze signal integrity and resolve I2C bus timing issues, cutting latency by 35%.

2. Senior Capstone Project with Draper

Systems & Hardware Engineer **Tools:** ESP32-S3, NVIDIA Jetson Orin Nano, I2S MEMS, System Architecture

- Developing an asymmetric two-tier event-driven acoustic-visual edge node for autonomous drone tracking and classification.
- Implementing continuous 1024-point FFT harmonic filtering on a Tier 1 ESP32-S3 and TDK InvenSense ICS-43434 I2S MEMS microphone.
- Designed a GPIO wakeup handshake with Tier 2 NVIDIA Jetson Orin Nano vision processor only upon valid acoustic triggers.

3. Pipelined RV32I Processor Core

RTL Designer **Tools:** SystemVerilog, OpenLane 2, Sky130, Verilator

- Microarchitected a 5-stage pipelined RISC-V (RV32I) core in SystemVerilog featuring hazard detection, data forwarding, pipeline stalls, and branch flushing.
- Integrated separate instruction and data memory interfaces mapped onto dedicated AHB-Lite Manager ports.
- Optimized total core power to 1.25 mW @ 95.2 MHz across 1,066 standard cells; validated pipeline state using SystemVerilog Assertion (SVA) testbenches.

4. AMBA AHB-Lite Bus Interconnect & Subordinate IP

RTL Designer **Tools:** SystemVerilog, Verilator, OpenLane 2

- Designed a reusable, parameterized AMBA AHB-Lite interconnect supporting dual-port SRAM memory subordinates with dynamic wait-state generation and multi-size access decoding.
- Authored formal SVA checkers enforcing ARM AHB-Lite protocol compliance, validating control/address/data stability during wait-states.

5. 63-Tap FIR Filter Hardware Accelerator

RTL & DSP Engineer **Tools:** SystemVerilog, AHB-Lite, DSP, Verilator

- Implemented a 63-tap symmetric FIR filter hardware accelerator with pre-addition folding and a pipelined MAC reduction tree.
- Integrated an AHB-Lite subordinate wrapper (‘ACCEL.sv‘) featuring custom Control and Status Registers (CSRs) and a 256-entry buffered output memory.
- Verified complete execution pipelines via a custom testbench simulating bulk coefficient loading, stream processing, and interrupt-driven status flags.

6. Bit Golf – FPGA VGA Game

Hardware/Software Co-Design Engineer **Tools:** VHDL, FPGA, VGA Graphics, Finite State Machines, System Integration

- Designed a modular VHDL-based VGA game engine on an UPduino FPGA rendering real-time 640x480 graphics via a 25.125 MHz pixel clock.
- Developed core game logic and multi-state finite state machines, implementing club selection rules and directional aiming mechanics.
- Built custom display overlay renderers for the ball, direction arrows, and club UI indicators, managing layout priority and compositing over the tile-based background.